Inequalities — problem 7.62

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Each of the vectors $\vec{a}_1, \ldots, \vec{a}_n$ has length ≤ 1. Prove that the signs can be chosen in the sum

$$\vec{c} = \pm\vec{a}_1 \pm \cdots \pm \vec{a}_n \quad \text{so that } |\vec{c}| \leq \sqrt{2}.$$